STRESS DETECTION WITH EEG USING TRANSFORMERS

Project-1 (EC47003) report submitted

To

Indian Institute of Technology Kharagpur

In partial fulfilment for the award of degree of

Bachelor of Technology

In

Electronics and Electrical Communication Engineering

By

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Under the supervision of

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Contents

Abstract

This report presents an account of my Bachelor Thesis Project (BTP)-1 at Electronics and Electrical Communication Department, Indian Institute of Technology Kharagpur under Prof. Dr. Basudev Lahiri. My main focus in this project is to build an effective model to use EEG Data in order to identify stress markers.

A series of experiments had been conducted on the data obtained from SAM-40 dataset of 40 subjects across 3 different tasks and a relaxed state. Features have been looked at from a statistic variability point of view to select for a good metric for stress. Delta/Alpha band power ratio was tried as such a metric of statistic relevance, but traditional classification techniques did not provide satisfactory results.

Later, more advanced models like LSTM and EEGNet had been used with better results but still not entirely satisfactory results with the models struggling to classify relaxed class because of severe class imbalance (3:1).

Transformers were utilized to be trained on a random set of 30 tasks across all the subjects. It provided a very good result of 91% accuracy overall and 79% relax metrics, the best until then. This pre trained Transformer was validated with the rest of data to be 95% accurate overall and had 89% relaxed precision and recall.

The Transformer model and all the rest of the experiments' code is hosted on Github at https://github.com/IshikiLucas/countdown/blob/9fc635251b24a358298c0a5e9a0ee403e1943e05/EggBurst2.ipynb

Acknowledgement

Foremost, I am incredibly grateful to Prof. Basudev Lahiri (Faculty Advisor and Project Guide), Prof. Tarun Kanti Bhattacharya (HoD, E&ECE) and Prof. Arun Chakraborty (Dean of Student Wellbeing) for providing me this project at E&ECE department of IIT Kharagpur. It is only because of their express support and encouragement, have I been able to do this internship successfully.

I am grateful to Ayush Tibrewal for hosting the data publicly and the team Rajdeep Ghosh et al for recording this data for the express use in stress-based diagnosis.

My sincere thanks to my seniors and mentors Subhanita Roy and Souvik Das for guiding me through difficult parts of the internship by thoughtful suggestions, insightful comments and corrections. I appreciate their helpful attitude towards me in my time of need.

I also thank everyone from the Institute fraternity who stepped up for me as their family.

Introduction

Electroencephalogram(EEG) is a non-invasive method to record the electrical signals inside the brain with (16 or 32) electrodes studded across the top of the head. It is generally used for medical diagnosis of disorders like epilepsy or seizures.

EEG is generally categorized by the frequency bands, namely Alpha(8-13Hz), Beta(14-30Hz), Delta(0.5-4Hz) and Theta(4-8Hz). Each of these frequency bands correspond to a certain kind of activity in the brain, be it relaxed state or thinking or doing some reflexive action, etc.

Stress or psychological stress is usually diagnosed by questionnaires rather than EEG. Although stress is a response from the brain that can make the individual's body to go into overdrive, brain's electric response is not usually used for its diagnosis.

The frequency bands are used sometimes to describe a disturbed state of mind as compared to a calm, relaxed state. This has been a good marker for stress from EEG across many papers.

There is a popular visual for stress in a lie detection setup, where their ECG is recorded. I wished to see if such a thing can be detected through EEG data, where a model could look at the EEG and assess if the subject was stressed or not.

For stress analysis, the subject is made to do some tasks(like Arithmetic problems) with variable stress induction and a rest state is taken as a contrast. Then, the data obtained from EEG recordings is used for prediction, analysis of stress.

Literature Review

I had gone through a few papers that tried to achieve the same.

Fares Al-shargie et al had achieved a 94.7% accuracy while using SVM with Error Correcting Code for penalty, trying to classify subjects' data into three levels of stress based on the verbal feedback given by them. They had depended upon the Spectral Power Density of frequency bands as input features. (Fares Al-shargie, 2017)

Valencia-Miranda Mireille et al had used Ratios of Band Powers (beta/alpha and theta/beta)as a metric to gauge stress while trying to analyse how playing a videogame 'Plants vs Zombies 2' can help with stress. (Valencia-Miranda Mireille, 2023)

The above papers had me interested in band powers and their ratios as possible features for stress classification.

But, Wenhui Cui et al had utilized a GPT model on specific encodings of raw EEG data from a TUH database with over 14,000 subjects as inputs to predict the masked encodings in what they refer to as, 'Neuro GPT'. (Wenhui Cui)

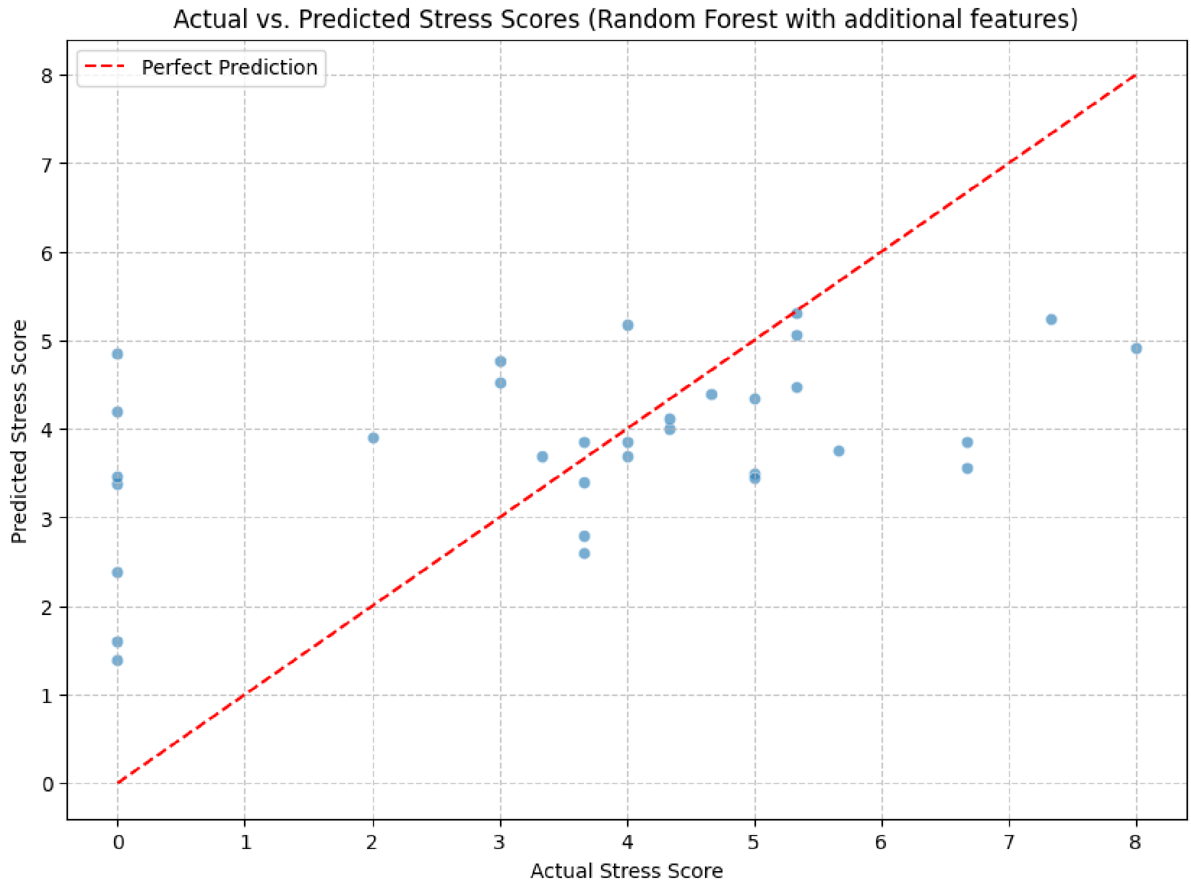
This inspired me to take a similar approach on the EEG data and use these encodings and how close they are in the hyperdimensional subspace to classify the data into stress and relaxed.

Data and methods

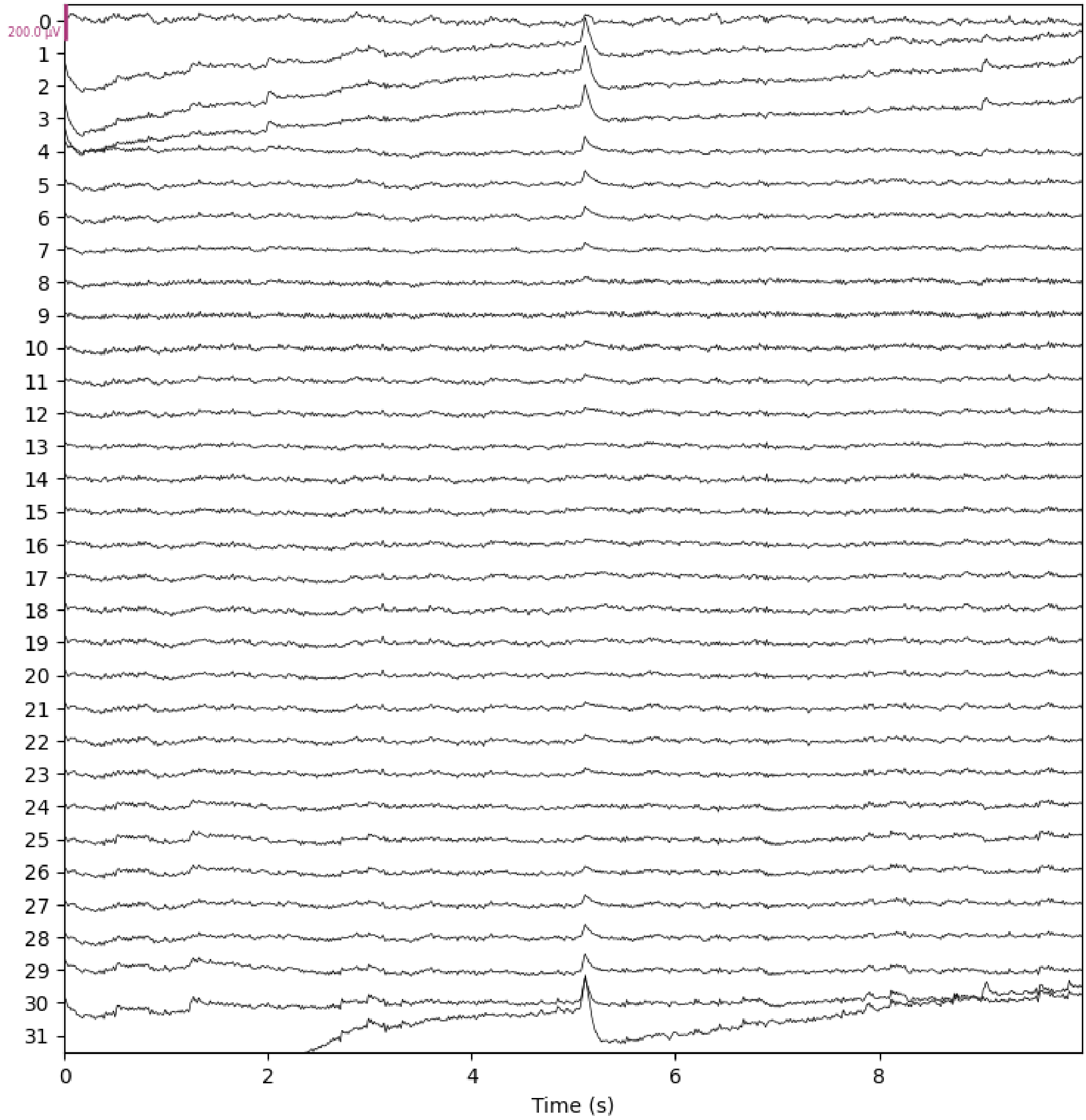
All the data used has been taken from SAM-40 EEG dataset made available publicly on Kaggle as ‘Raw EEG Stress Dataset’ by Ayush Tibrewal.

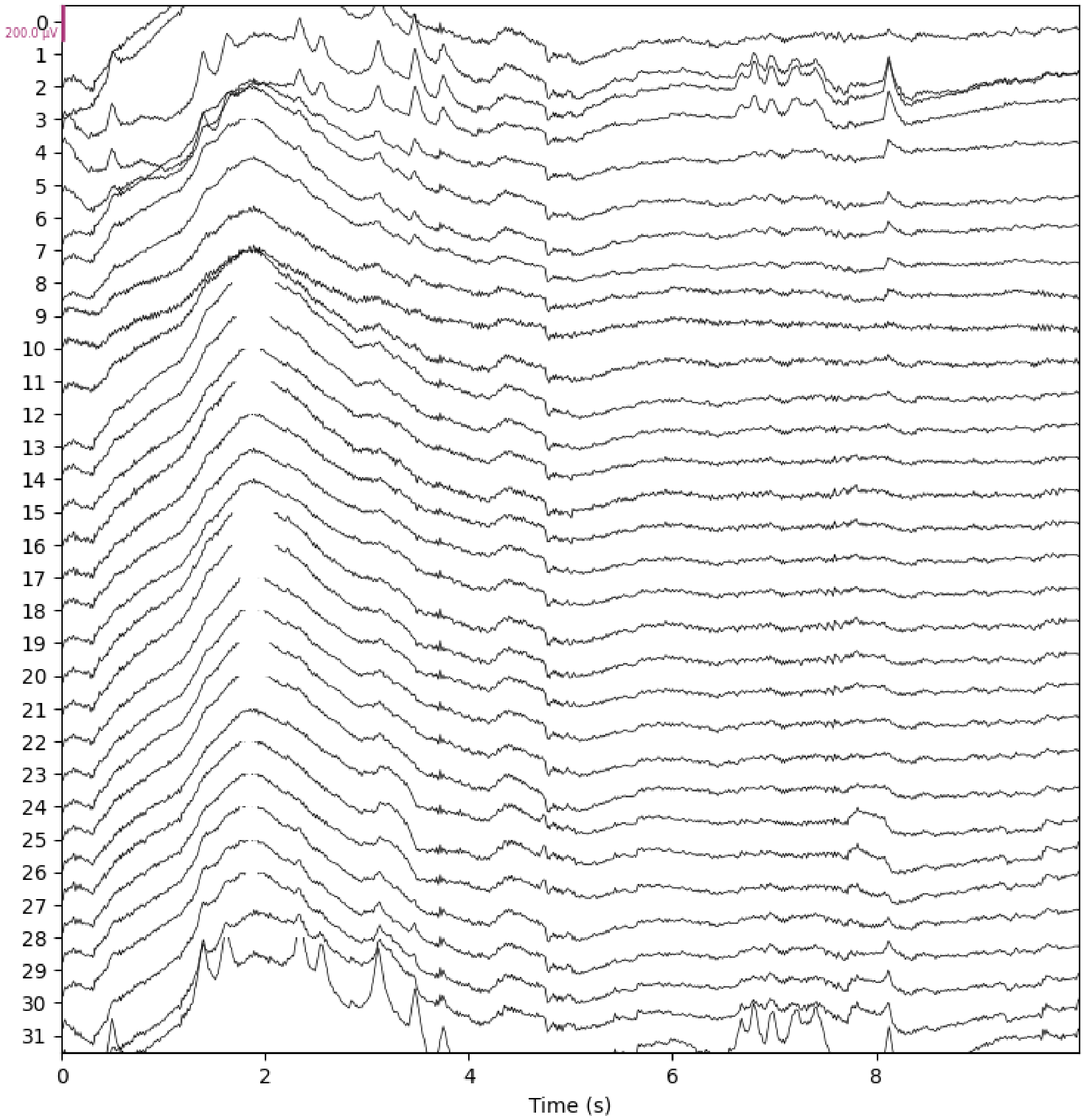
The data has been recorded by (Rajdeep Ghosh, 2022) across 40 subjects (mean age:21.5) as they were doing 4 different tasks, namely solving Arithmetic problems, Stroop Colour Word test, identifying Mirror images and relaxed state.

The EEG readings were taken by Rajdeep Ghosh et al using 32-channel Emotiv Epoc Flex gel kit for the purpose of stress evaluation. They also included a table of verbal feedback for how stressed each subject felt during an activity (other than relaxed). Although the feedback could help us with classification with a stress score, the verbal feedback is entirely subjective and was shown to not be of much statistic variation to be of much use in the classification.



So, I had just taken the EEG readings and ignored the verbal feedback for the classification.




200.0 µV
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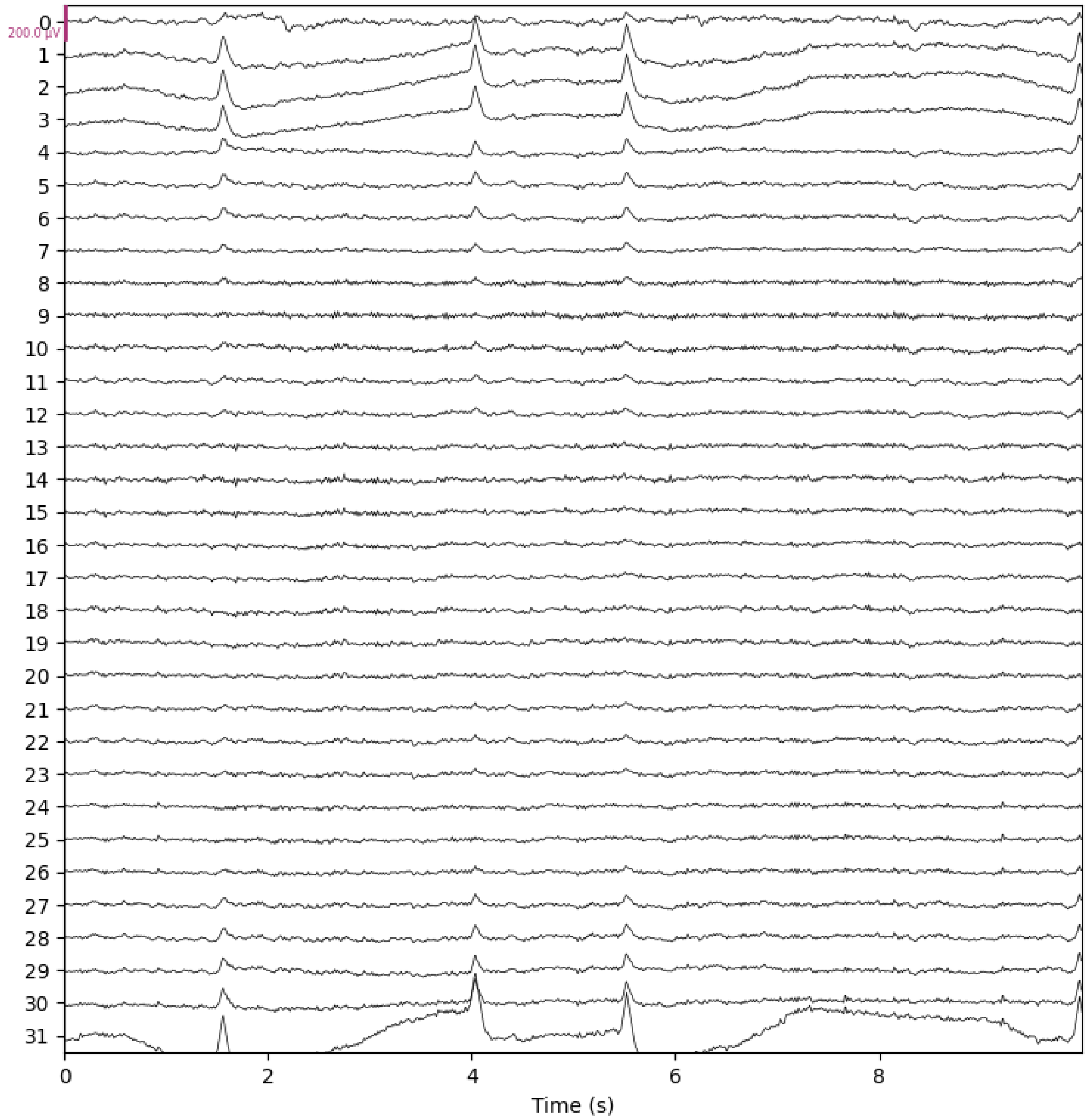
Time (s)



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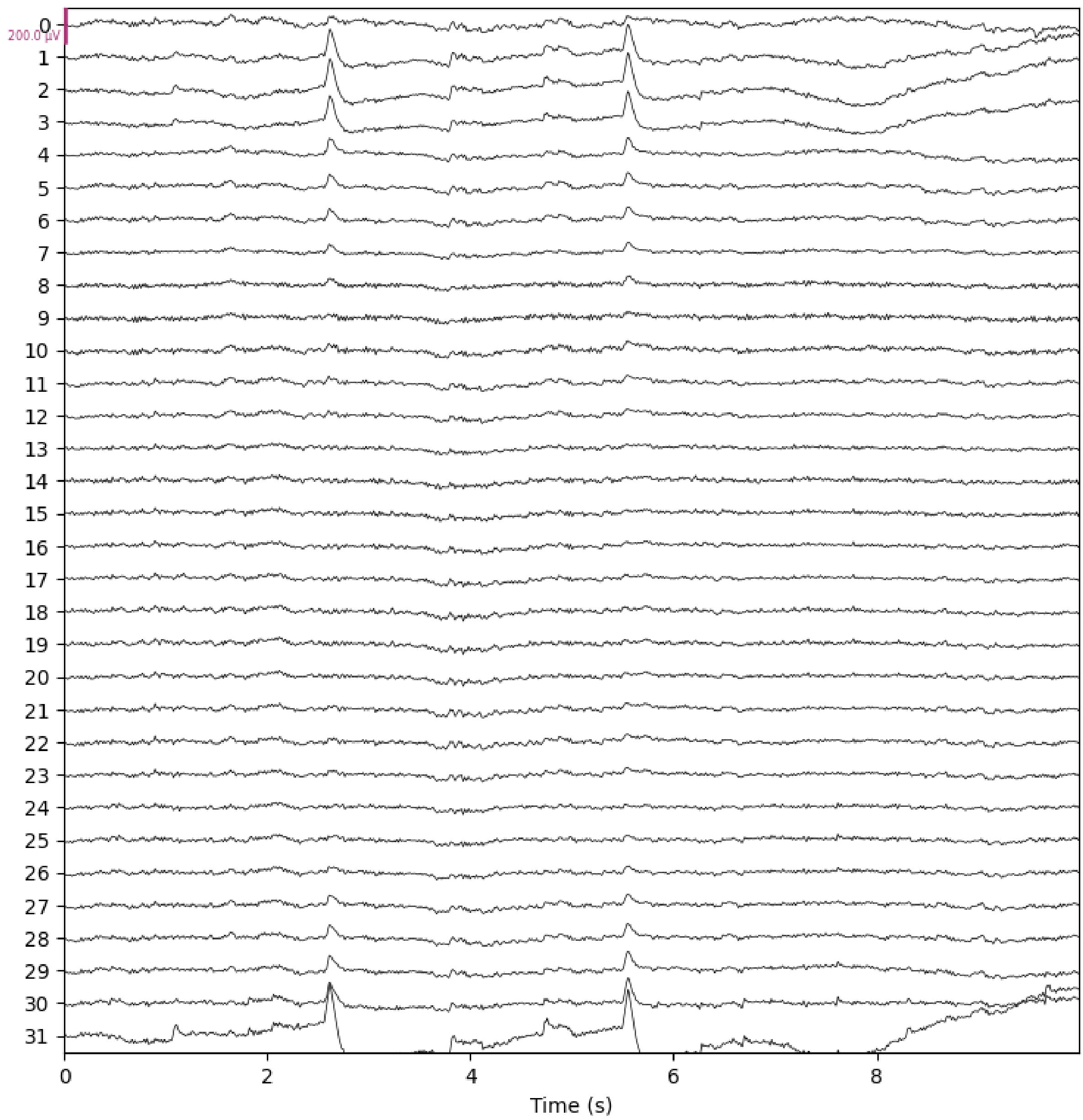
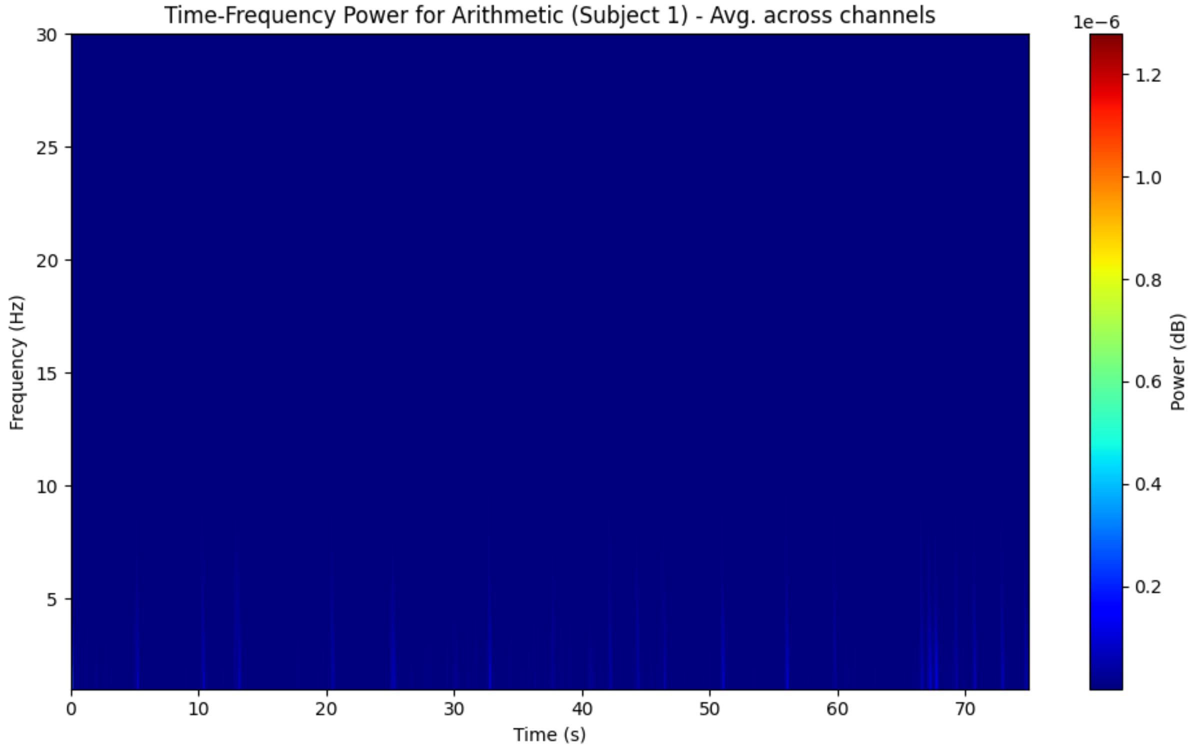
Time (s)

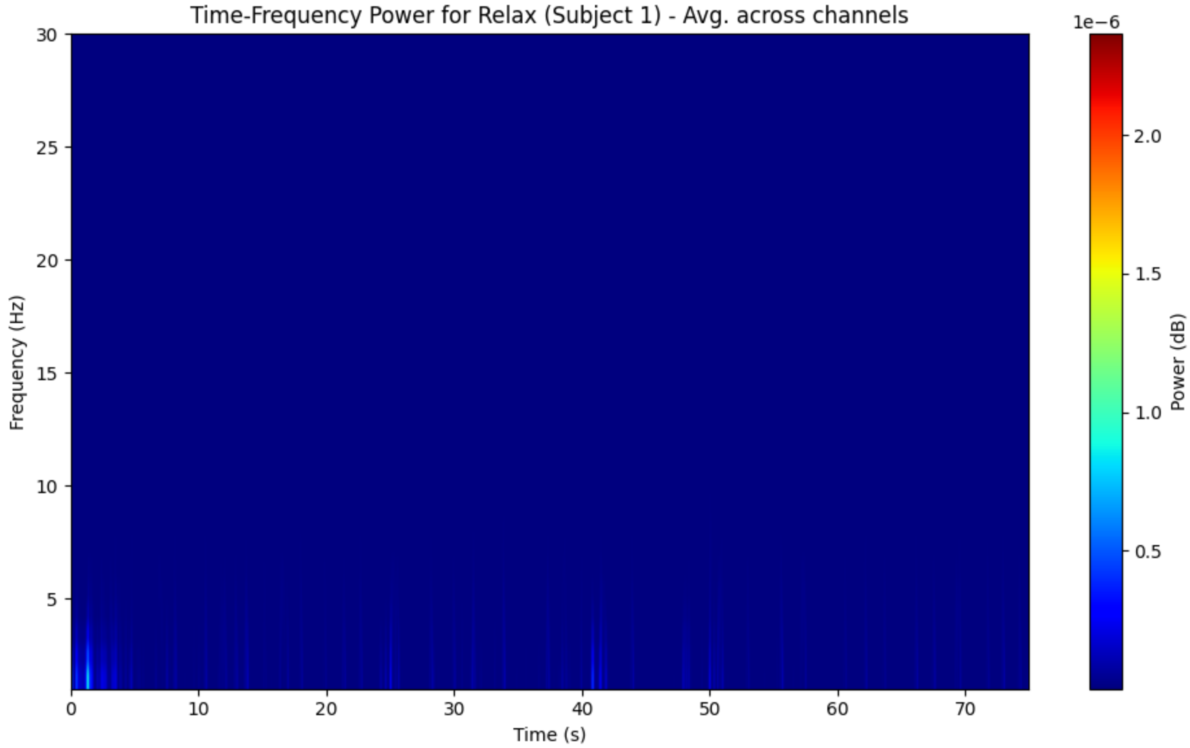


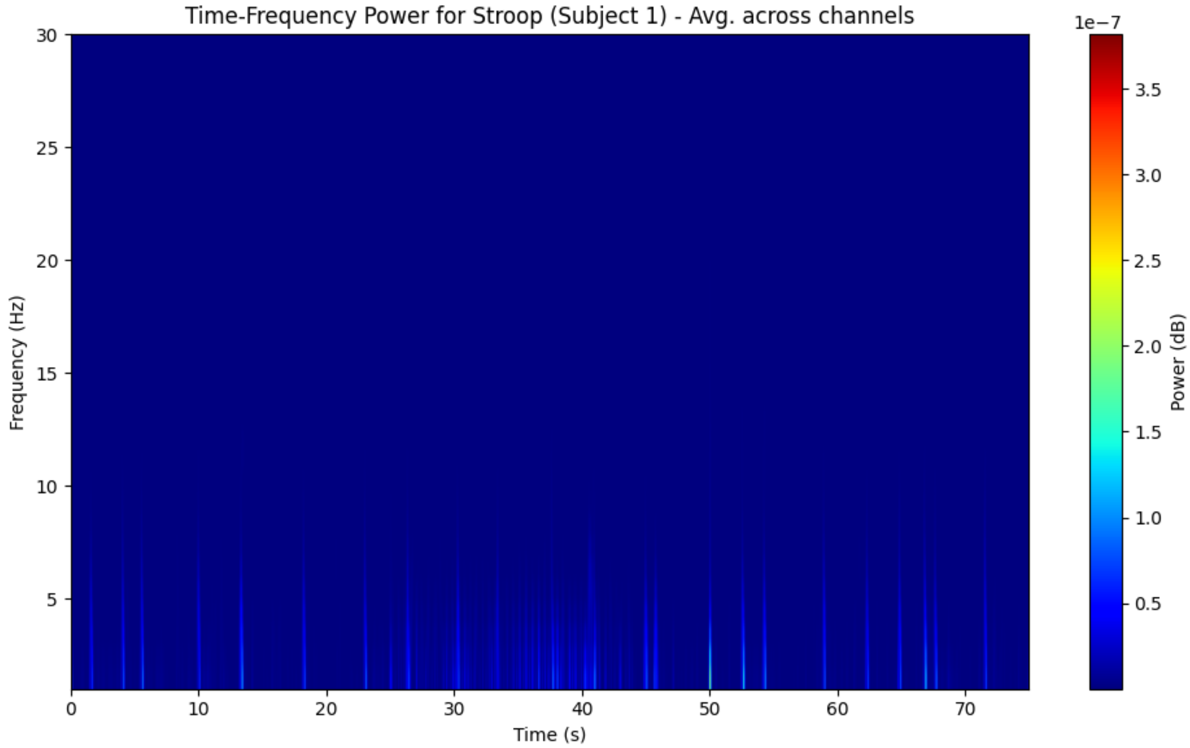
EEG Reading of Sub-1 performing RELAX, STROOP, ARITHMETIC, MIRROR

Statistical Analysis

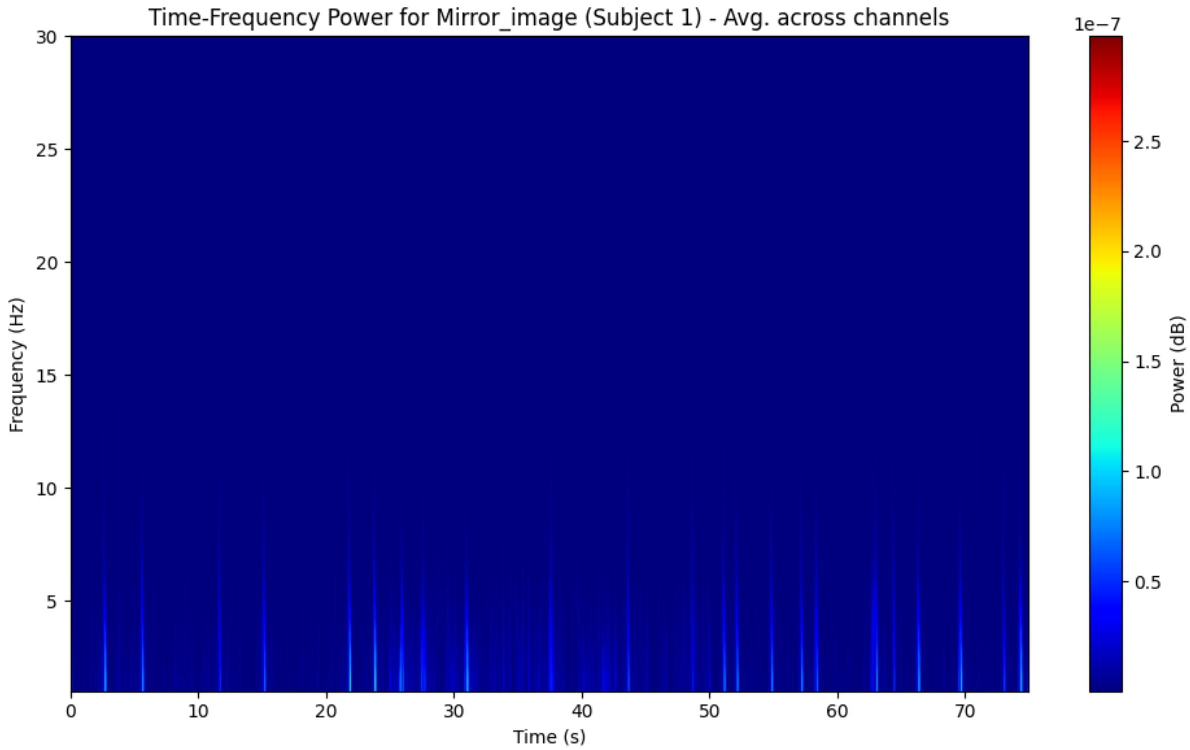
I had normalised the data to remove any additional artifacts. This data was then used to produce Discrete Wave Transform and band powers (PSD) of alpha, beta, theta and delta band frequencies to check statistic significance and usability in the model.



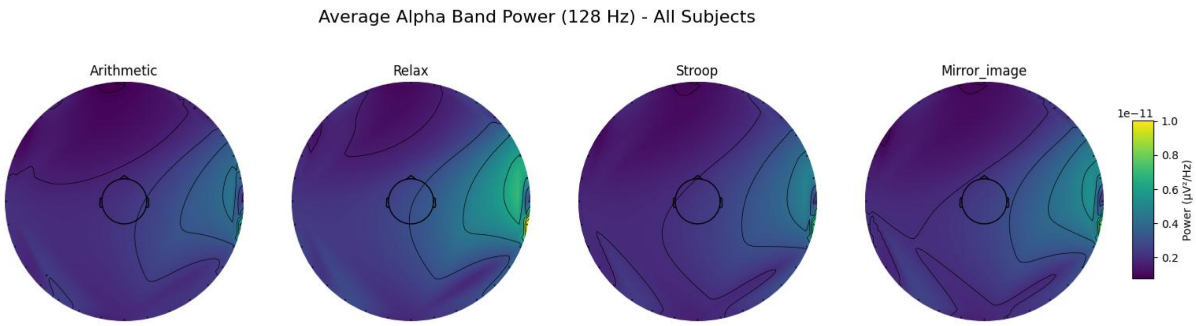


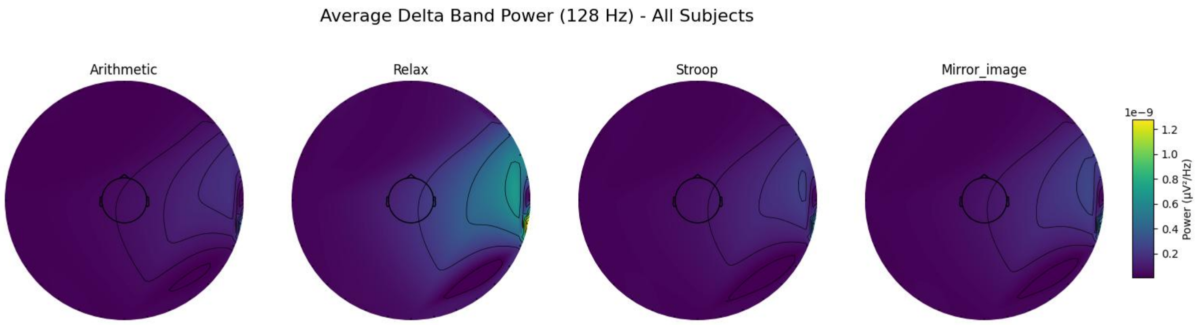


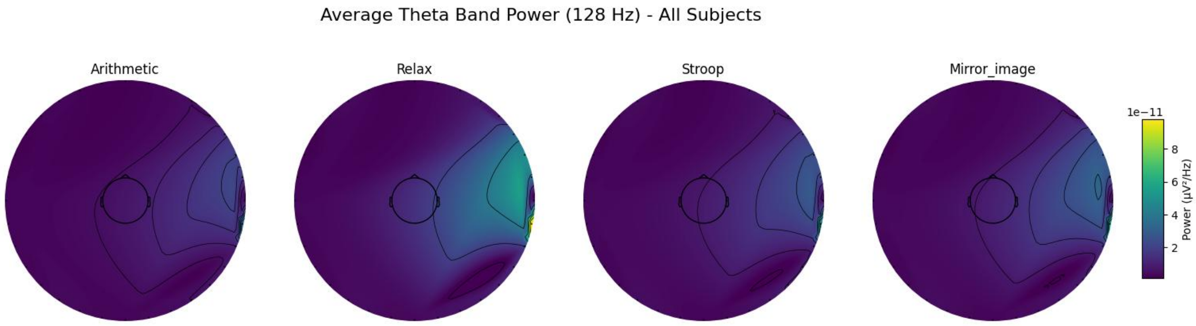
Time-Frequency Power for Stroop (Subject 1) - Avg. across channels
Frequency (Hz)
Time (s)
1e−7
Power (dB)
0 10 20 30 40 50 60 70
5 10 15 20 25 30
0.5 1.0 1.5 2.0 2.5 3.0 3.5

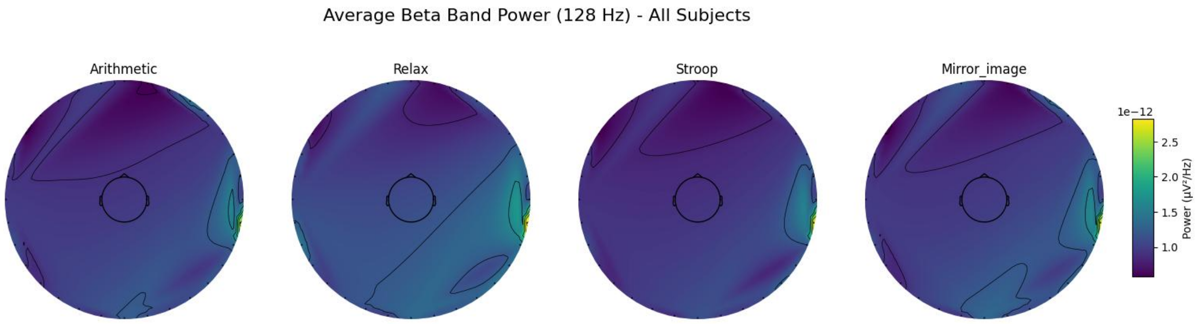


Time-Frequency Power for Mirror_image (Subject 1) - Avg. across channels
Frequency (Hz)
Time (s)
1e−7
Power (dB)
0 10 20 30 40 50 60 70
5 10 15 20 25 30
0.5 1.0 1.5 2.0 2.5









On analysing subject wise, I had found that delta and alpha bands show noticeable variation.

Performing ANOVA for each frequency band across tasks (128 Hz data):

--- Delta Band ---	F-statistic: 5.927
P-value: 0.001	Significant difference found in Delta power across tasks.

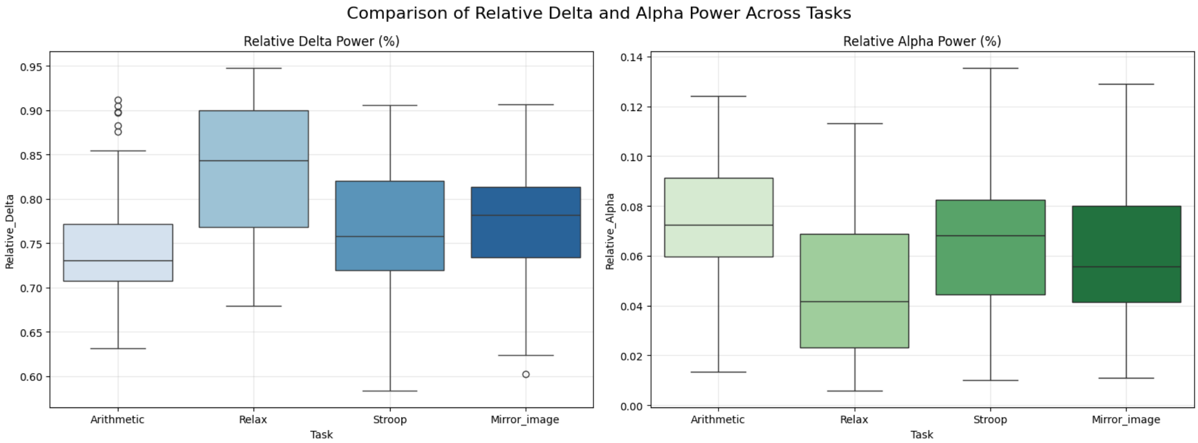
--- Theta Band ---	F-statistic: 2.457
P-value: 0.063	No significant difference in Theta power.

--- Alpha Band ---	F-statistic: 3.775
P-value: 0.011	Significant difference found in Alpha power across tasks.

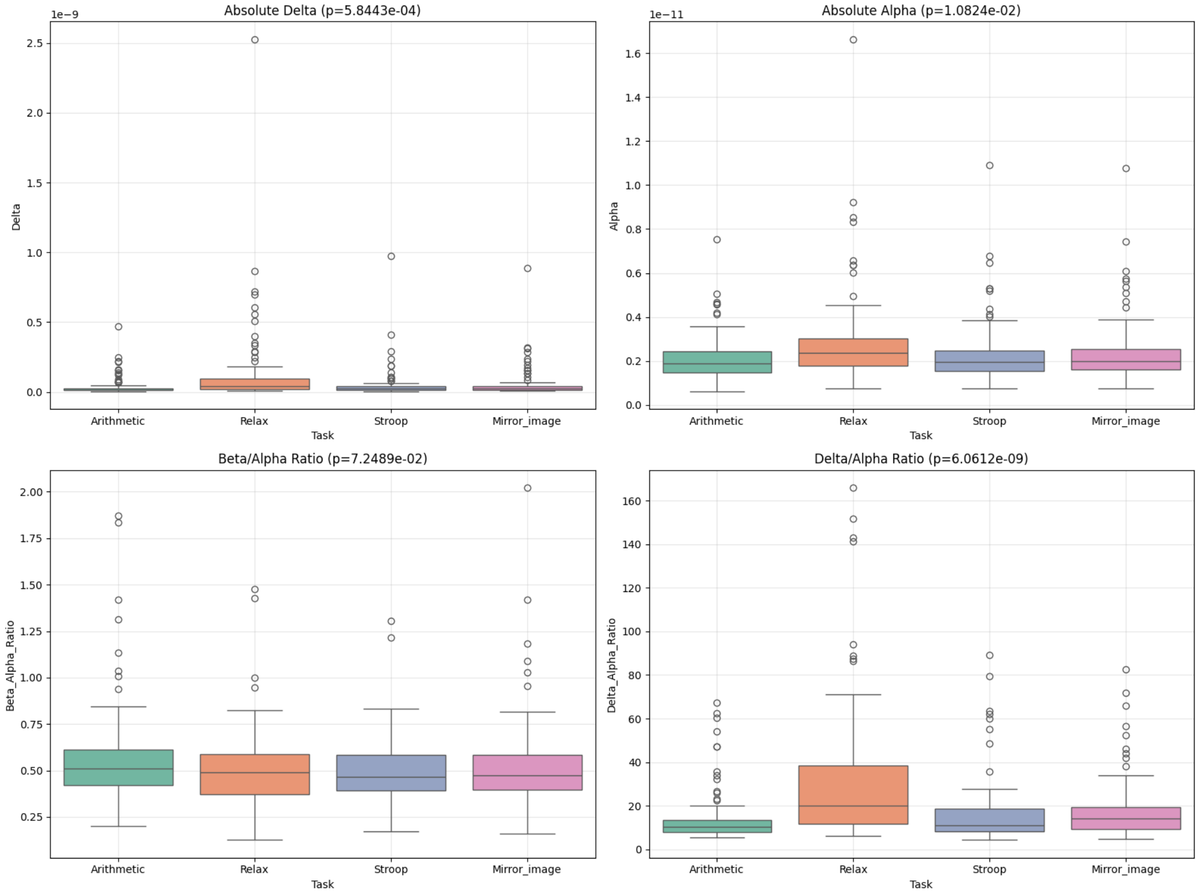
--- Beta Band ---	F-statistic: 2.176
P-value: 0.090	No significant difference in Beta power.

A P-Value less than 0.05 shows statistical significance.

Probability value gives us the probability with which you can observe any change without that component actually contributing to said change. So, there's only 1% chance that alpha band would not contribute to any observable change in the accessible data.



Since alpha and delta bands show such variability I had taken delta/alpha as a metric to try and classify the data.



As we can see the P value of del/alp is quite low making it statistically significant.

Experiments

I had used regression models to classify stress and relaxed states. I quickly ran into issues as my stress accuracy was pretty low.

--- Full Dataset Model Results (Relax vs. Stress) ---

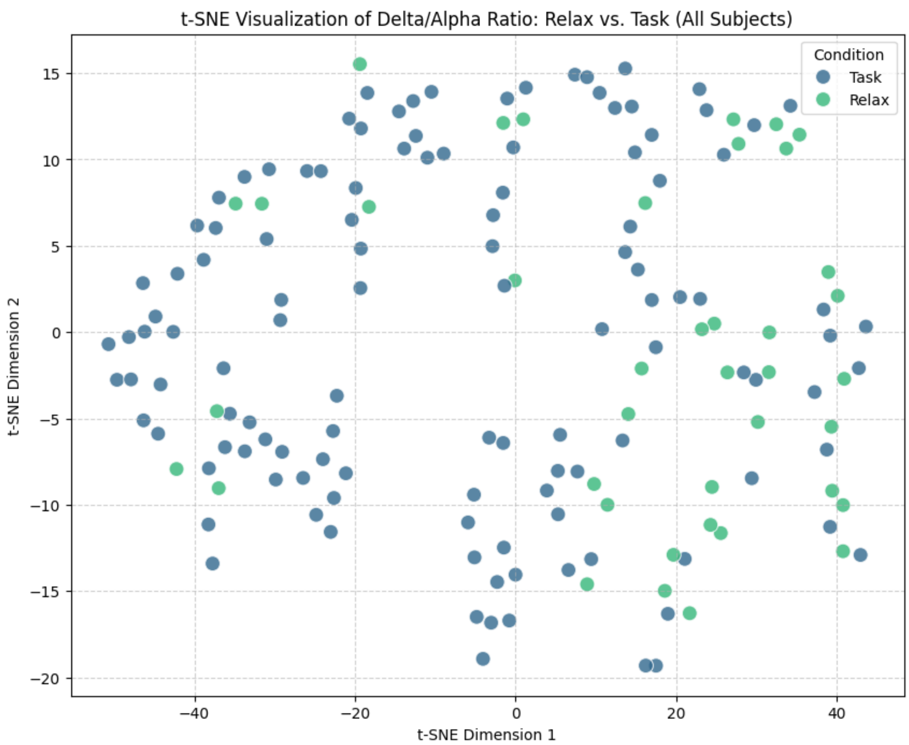
	precision	recall	f1-score	support
Task	0.87	0.78	0.82	120
Relax	0.49	0.65	0.56	40
accuracy			0.74	160
macro avg	0.68	0.71	0.69	160
weighted avg	0.77	0.74	0.75	160

Precision=TP/(TP+FP) Accuracy=(TP+TN)/(TP+TN+FP+FN)

Recall=TP/(TP+FN)

F1-score=2(Precision)(Recall)/(Precision+Recall)

So, I had conducted a t-Scholastic Neighbour Embedding test to see the kind of variability I am dealing with.



It showed a very complex variability pattern, where many stress markers are right there with the relaxed ones. It comes as no surprise then, that the model using simple regression techniques was confusing relaxed for stress data.

So I trained the data on an Multi Layer Pereptron (MLP) to see if that can improve on the stress recall issue.

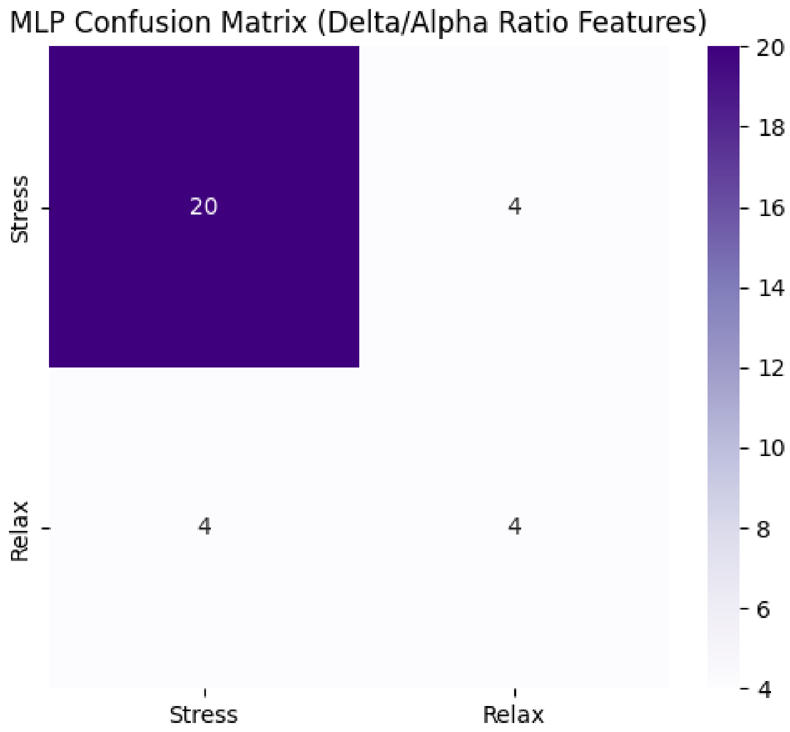
```
--- MLP Raw Data Classification Report ---
              precision    recall  f1-score   support

 Stress Task       0.84      0.94      0.89       450
       Relax       0.72      0.48      0.58       150

    accuracy                           0.82       600
   macro avg       0.78      0.71      0.73       600
weighted avg       0.81      0.82      0.81       600
```

There still was not much improvement. Our baseline random guess would have a probability of 25%. It is better than 25% but still not useable for classification.

So, I tried to augment the data, assuming there wasn't enough data to go around for the model to find out features, effectively doubling the amount of data and trained it again on MLP and also implemented a penalized class weight system to outgo the class imbalance of 3:1.

```
--- MLP Augmented Data Classification Report ---
              precision    recall  f1-score   support

 Stress Task       0.88      0.92      0.90       894
       Relax       0.73      0.63      0.68       298

    accuracy                           0.85      1192
   macro avg       0.80      0.78      0.79      1192
weighted avg       0.84      0.85      0.85      1192
```

```
-- MLP (Delta/Alpha Ratios) Classification Report ---
             precision    recall  f1-score   support

      Stress       0.83      0.83      0.83        24
       Relax       0.50      0.50      0.50         8

    accuracy                           0.75        32
   macro avg       0.67      0.67      0.67        32
weighted avg       0.75      0.75      0.75        32
with penalised class weights on normalised data
```



This gave a better result but I had wished to better this metric even more.

So, I changed my strategy and used EEGNet, a legacy library for analysis of EEG signals to see if any classification can be done using normalized EEG Data alone, assuming del/alp does not have enough features for training.

--- EEGNet Classification Report ---

	precision	recall	f1-score	support
Stress	0.79	0.95	0.86	3576
Relax	0.61	0.22	0.32	1192
accuracy			0.77	4768
macro avg	0.70	0.59	0.59	4768
weighted avg	0.74	0.77	0.73	4768

EEGNet did not give the best results infact the stress recall is worse than logistic regression.

Then I implemented LSTM to see if there were any features that need to considered in tandem to improve the metrics. I had observed 83% accuracy, highest so far, yet still struggling with relax identification.

--- LSTM Classification Report ---

	precision	recall	f1-score	support
Stress Task	0.85	0.93	0.89	3576
Relax	0.72	0.52	0.61	1192
accuracy			0.83	4768
macro avg	0.79	0.73	0.75	4768
weighted avg	0.82	0.83	0.82	4768

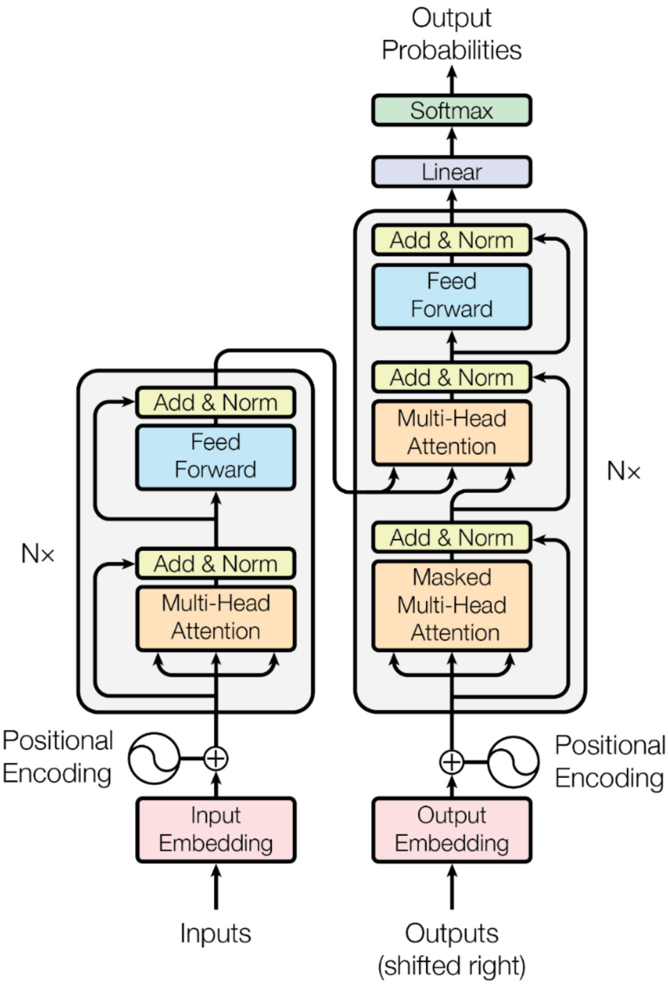
Transformers

Transformers are a part of the revolutionary step that makes Large Language Models like Chat-GPT possible. ‘Attention is all you need’, a paper by (Ashish Vaswani, 2017) introduces the concept of self-attention which uses Query, Key of dimension d_k and Value of dimension d_v to determine the position of a certain input encoding in a hyperdimensional space, thereby placing similar concepts or tokens as they are called, near each other.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$ (Ashish Vaswani, 2017)

Attention here is called Scalar Dot-Product Attention. d _k is the dimension of keys(K).

Transformers were originally used for language translation, but have found uses in many fields, AI being the most popular. I was prompted to employ Transformers in this project by Wenhui Cui et al, where they had used transformers to classify data from EEG. They called it Neuro-GPT. This led me to try and build a type of Neuro-GPT that can effectively classify Stress and Relaxed data from a small dataset, assuming that other models I had tried with could not get much useful features from the amount of data available.



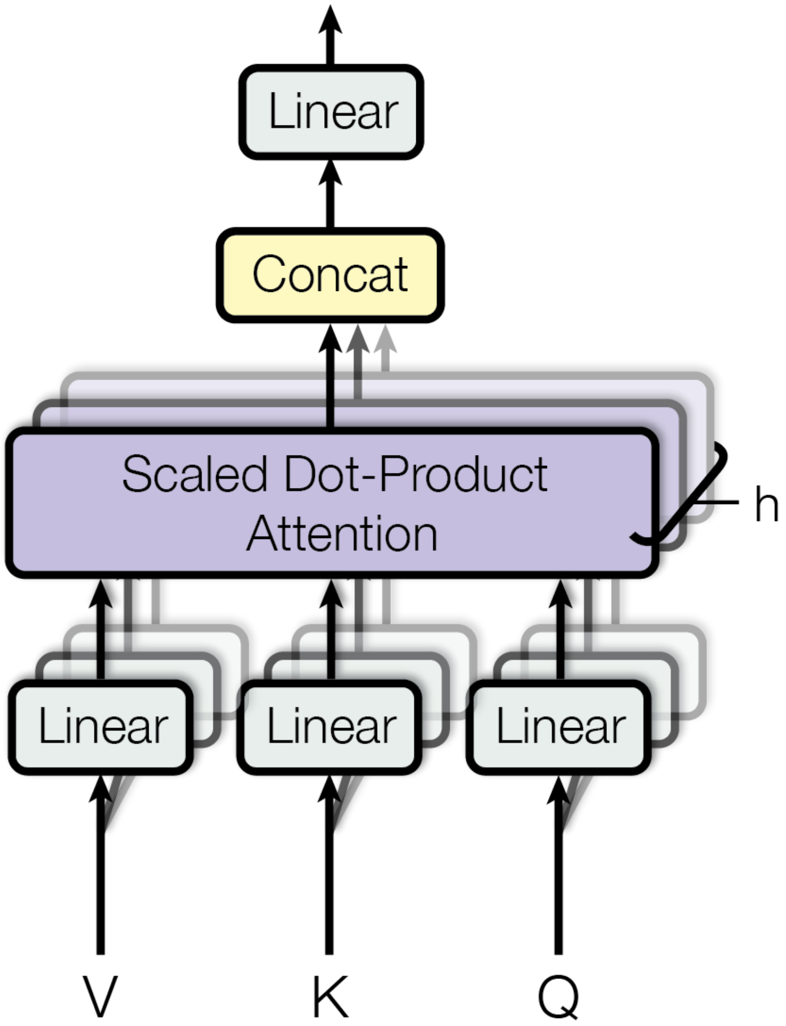
Transformer Model (Ashish Vaswani, 2017)

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this.

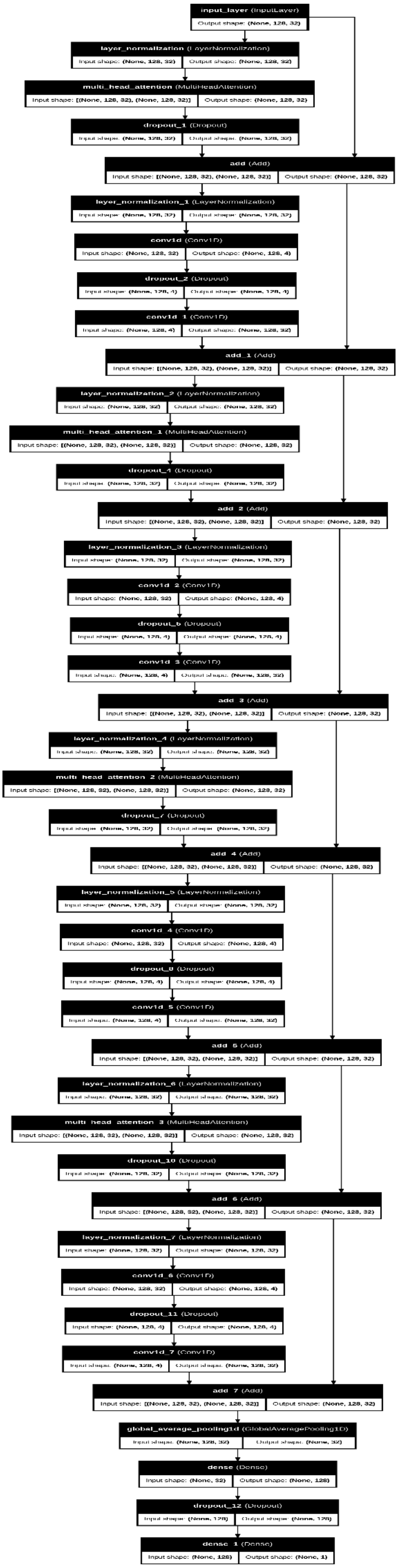
$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head_1}, \ldots, \mathrm{head_h})W^O$$

$$\mathrm{where\,head_i} = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$. (Ashish Vaswani, 2017)



Multihead Attention (Ashish Vaswani, 2017)



input_layer (InputLayer)
Output shape: (None, 128, 32)
layer_normalization (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
multi_head_attention (MultiHeadAttention)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
dropout_1 (Dropout)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
add (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_1 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
conv1d (Conv1D)
Input shape: (None, 128, 32)
Output shape: (None, 128, 4)
dropout_2 (Dropout)
Input shape: (None, 128, 4)
Output shape: (None, 128, 4)
conv1d_1 (Conv1D)
Input shape: (None, 128, 4)
Output shape: (None, 128, 32)
add_1 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_2 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
multi_head_attention_1 (MultiHeadAttention)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
dropout_4 (Dropout)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
add_2 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_3 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
conv1d_2 (Conv1D)
Input shape: (None, 128, 32)
Output shape: (None, 128, 4)
dropout_5 (Dropout)
Input shape: (None, 128, 4)
Output shape: (None, 128, 4)
conv1d_3 (Conv1D)
Input shape: (None, 128, 4)
Output shape: (None, 128, 32)
add_3 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_4 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
multi_head_attention_2 (MultiHeadAttention)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
dropout_7 (Dropout)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
add_4 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_5 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
conv1d_4 (Conv1D)
Input shape: (None, 128, 32)
Output shape: (None, 128, 4)
dropout_8 (Dropout)
Input shape: (None, 128, 4)
Output shape: (None, 128, 4)
conv1d_5 (Conv1D)
Input shape: (None, 128, 4)
Output shape: (None, 128, 32)
add_5 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_6 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
multi_head_attention_3 (MultiHeadAttention)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
dropout_10 (Dropout)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
add_6 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
layer_normalization_7 (LayerNormalization)
Input shape: (None, 128, 32)
Output shape: (None, 128, 32)
conv1d_6 (Conv1D)
Input shape: (None, 128, 32)
Output shape: (None, 128, 4)
dropout_11 (Dropout)
Input shape: (None, 128, 4)
Output shape: (None, 128, 4)
conv1d_7 (Conv1D)
Input shape: (None, 128, 4)
Output shape: (None, 128, 32)
add_7 (Add)
Input shape: [(None, 128, 32), (None, 128, 32)]
Output shape: (None, 128, 32)
global_average_pooling1d (GlobalAveragePooling1D)
Input shape: (None, 128, 32)
Output shape: (None, 32)
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Input shape: (None, 32)
Output shape: (None, 128)
dropout_12 (Dropout)
Input shape: (None, 128)
Output shape: (None, 128)
dense_1 (Dense)
Input shape: (None, 128)
Output shape: (None, 1)

Total params: 542,737 (2.07 MB)

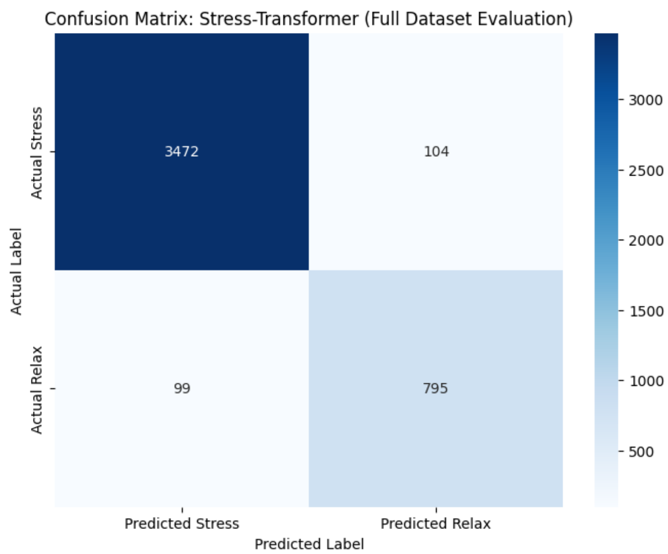
Trainable params: 542,737 (2.07 MB)

Given the number of parameters, to train the whole 140 task dataset is too much for my device. So, I decided to start small with 30 random tasks to train the model instead and it had produced one of the best results so far.

```
- Transformer (Stress-GPT: 30 Session Subset) Classification Report -
              precision    recall  f1-score   support

      Stress       0.94      0.95      0.94       715
       Relax       0.78      0.78      0.78       179

    accuracy                           0.91       894
   macro avg       0.86      0.86      0.86       894
weighted avg       0.91      0.91      0.91       894
```

78% recall and precision for stress is unprecedented. There is the possible risk that the model has just overfitted to the available data only. So I had used the rest of the data(all 140 tasks) to validate the model.

```
-- Full Dataset Evaluation Report ---
              precision    recall  f1-score   support

      Stress       0.97      0.97      0.97      3576
       Relax       0.88      0.89      0.89       894

    accuracy                           0.95      4470
   macro avg       0.93      0.93      0.93      4470
weighted avg       0.95      0.95      0.95      4470
```



The result came as a pleasant surprise. 95% accuracy overall and 89% relaxed precision is exceptional given our class imbalance.

Results and Discussion

While using delta/alpha ratio as the metric for stress in EEG

Model	Overall Accuracy	Relax F1 Score
Regression(del/alp)	74%	56%
MLP(low data)	82%	58%
MLP(aug data)	85%	68%
MLP(class weights)	75%	50%

While using EEG Data directly to train more complex models

EEGNet	77%	32%
LSTM	83%	61%
Transformer(30 tasks)	91%	78%
Transformer(full data)	95%	89%

Even if the model had marked all the relaxed states as stress, we would have 75% accuracy(given the class imbalance). So that is the standard baseline I sticked to for a sense of satisfactory result.

As can be observed, just a little more than a half of these models yield any satisfactory result above the 75%. Transformer outperformed all the rest in every metric with 95% accuracy, proving the superiority of the method in cases with high class imbalance and relatively less amount of data to train on, as it only used 30 random tasks to learn about the behaviour of the whole dataset.

I believe that Transformers' attention that places the features that are related to each other in the position encoding proved powerful when classification.

In the future, transformers could be used to train on huge datasets like the TUH used by Neuro-GPT to further technologies like BCI and to increase efficiency of diagnostic techniques.

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